

[Collaborative Docs](#) >

Discussion Agenda 06/29/2016 (Isolated Disk + AGORA Paper 4)

Below is a discussion agenda / live progress report for the 9th Web Conference of the AGORA Project (Jun. 29, 9 am PDT/6 pm CEST). This telecon will particularly focus on the progress and future work for AGORA Paper 4 and planning of the upcoming 5th AGORA Workshop. Some items below were discussed among some members of the Steering Committee on Jun. 15 in preparation of this Group-wide meeting.

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Check out the important items in red color below!

(1st revision: 06/30/2016)

[1. Progress Report \(as of Jun. 2016\)](#)

We are actively making progress, chiefly, in three different areas below (numbered tentatively).

	PAPER "2"	PAPER "3"	PAPER "4"																																																				
GOALS	• Run Proof-of-concept-DM (~1e11 Ms halo) at higher resolution	• Implement and test common astrophysics package • Run Proof-of-concept-NOSUBGRID (~1e11 Ms halo)	• Calibrate SF/Fbck recipes using common disk ICs																																																				
LEADERS	• Oliver Hahn (OCA, Nice) • Christine Simpson (HITS)	• Nick Gnedin (Fermilab) • Piero Madau (UCSC) • Britton Smith (Edinburgh)	• Oscar Agertz (Surrey) • Romain Teyssier (Zurich) • James Wadsley (McMaster)																																																				
PARTICIPANTS	• Working Groups engaged: - III (common cosmo ICs) - IV (common analysis) - VI (dwarf galaxies) • Code representatives:	• Working Groups engaged: - I (common physics) - III (common cosmo ICs) • Code representatives:	• Working Groups engaged: - II (common disk ICs) - IV (common analysis) - V (subgrid physics) • Code representatives:																																																				
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PROGRESS MADE OR BEING MADE	• Please visit Paper 2's Workspace where Christine Simpson, Oliver Hahn and others have been updating the progress. • We will plan to have 5-code comparison. As of Jun. 2016, 2+ participating groups have already completed and submitted the production-level results (~100 pc resolution) to NERSC.	• Either via Britton's original interface or through Nick's API, GRACKLE is now fully implemented (equilibrium cooling version) in all 9 participating codes. • The resulting cooling functions is compared in isolated disk runs, Paper "4" (either in cooling only + noSF mode, or in cooling + SF + ThermalFbck mode). • Code groups such as ART-I and GIZMO have carried out cosmological simulations with various AGORA ICs, even including sophisticated subgrid physics.	• Please visit Paper 4's Workspace where Oscar Agertz, Romain Tessier and others have been updating the progress. • All 9 participating code groups have updated their "first" datasets (i.e. Grackle + noSF run) since the last Workshop. The latest composite images for the "first" datasets on their Workspace were created on May 5, 2016. • All 9 participating code groups have submitted their "second" datasets (i.e. Grackle + SF + ThermalFbck run), although 2+ groups are revising their dataset as of Jun. 2016. The latest composite images for the "second" datasets on their Workspace were created on Jun. 18, 2016. • The agora-analysis-script repository is being accordingly updated as the analysis progresses.																
OTHER LINKS	• Please see the progress report in Paper 2's Workspace and/or in WG III's Workspace. • MUSIC Google group	• GRACKLE Google group • GRACKLE download & documentation • AGORA API for GRACKLE by Nick Gnedin	• Please see the progress report in Paper 4's Workspace and/or in WG V's Workspace. • yt-AGORA Google group																

[2. Paper Group 4 - Plan For The Next 2 Months Before 2016 Summer Workshop](#)

- Below are abbreviated talking points only for this particular telecon only. The writings below may not be updated as the ideas in Paper "4" evolve in time after the telecon. **Therefore, participants of Paper "4" are STRONGLY recommended to check out and follow the up-to-date consensus in the [Paper 4 collaborative doc](#).**

(1) June 29: GROUP-WIDE WEB CONFERENCE

(1-1) Discussion Item #1 – Purpose and Scope of Paper 4

- Let us focus on comparing different numerical schemes on an idealized settings in various dimensions such as angular momentum, star formation, clumps, etc. Our collected datasets already give a lot to talk about and a unprecedented ground for fair comparison.
- To expedite the progress of Paper, we may leave the feedback calibration for future papers (e.g. calibration against observations using metrics such as galactic winds, fountains, or main sequence SF). We will also fall back to the current 80 pc resolution runs, instead of 40 pc runs originally planned. [---> Both items agreed. In the Paper, however, we may discuss a list of possible calibration targets for future purpose.]

(1-2) Discussion Item #2 – Additional Analyses

- Today is the "last" chance to add additional diagnostics, if any. [---> We discussed the following additional items, some of which may be included in the Paper. Ji-hoon will take a look at some of these in the coming weeks. Your help will be welcomed in any of the above analysis items.]
 - Flared disk elevation map (average z along lines of sight on a x - y plane; e.g. Levine et al. 2006)
 - Time evolution of finest resolution element size ($[M/\rho]^{1/3}$ for SPH codes)
 - Radius vs gas disk average vertical height ($\sim$ thickness)
 - Radius-velocity dispersion (e.g. Tamburro et al. 2009)
 - Star formation rate surface density
 - Kennicutt-Schmidt relation
 - Clumps statistics (gas and stars; e.g. Mayer et al. 2016, Guo et al. 2014)
 - Fourier modes (e.g. Chakrabarti & Blitz 2009)

- We will try different definitions to find the centers of galaxies. We will also be careful in visualizing the data, especially SPH's, by testing and using different smoothing schemes. FYI, also check out [the analysis shopping list in the Collaborative Doc](#) made in 2015.

(1-3) Discussion Item #3 – Paper Draft

- Please submit short paragraphs introducing each participating codes, particularly focusing on the following three items: (1) hydrodynamics, (2) refinement or softening/smoothing strategy, and (3) how feedback scheme was implemented. Leave out details that were already included in the Flagship Paper, or the commonly adopted parameters across platforms such as the star formation prescription (see the [Collaborative Doc](#) for already discussed common runtime parameters). [---> Agreed. Please submit these paragraphs by July 15, 2016]
- July 15 is also the proposed deadline to make any revision to your submitted datasets. Afterwards, revisions to your dataset may be discouraged. [---> Agreed.]
- Let us start writing up obvious parts of the paper first such as the Introduction, while collecting materials from each groups. Versions of draft will be circulated as were done in the Flagship Paper. [---> Agreed. A core writing group formed: Kim, Nagamine, Primack, Teyssier]

(2) June 30 – July 15: ADDITIONAL ANALYSIS, INITIAL PARAGRAPH SUBMISSION

- Complete additional analyses. Submit code introduction paragraphs per each group. Last chance to make any revision to the submitted datasets.

(3) July 15 – August 12: ADDITIONAL ITERATION

- Another group-wide web conference in late July, if need be.

(4) August 13–14: 5TH WORKSHOP WORKING SESSIONS

- Interpret all the diagnostics compiled so far, and write up paragraphs on the spot. Further revision to an already submitted dataset will not be allowed at this point.

(5) Late August to Early September

- Submit the Paper by the end of summer.

3. Other Discussion Items**(1) Announcement of the 2016 Santa Cruz Galaxy Workshop, and Planning of the 5th AGORA Workshop**

- The annual [Santa Cruz Workshop on Galaxy Formation](#), hosted by Avishai Dekel and Joel Primack, will take place this summer at UC Santa Cruz on August 8–12, 2016. The concluding day (Friday) of the Galaxy Formation Workshop will also be the first day of the [5th AGORA Workshop](#), August 12–14, 2016.

4. Recaps**(1) AGORA SMBH Prescription (Accretion and Feedback):**

Please visit [this link](#).

(2) Strategy to Store Simulation Outputs:

Please visit [this link](#).

(3) yt–Sunrise Pipeline for AGORA on NERSC:

Please visit [this link](#).

(4) Mini–workshop Opportunities:

Mini–workshop for individual paper groups to write papers: KIPAC@Stanford is willing to provide funding opportunities for working group leaders to organize mini–workshops (~10 people). An example was the [yt–AGORA mini–workshop](#) at UCSC in Mar. 2014.

(5) Useful Links:

- Quick links to the important pages in AGORA Workspace, including internal policies: [here](#)
- Internal policy regarding the disclosure of AGORA simulation results: [here](#)
- Collection of softwares relevant to AGORA effort: [here](#)

[5. Your Feedback](#)

Let us hear anything AGORA, including the scientific and organizational effort currently being made. Voice your opinion during the web conference! Feel free to share your thoughts in the comments section below, or on the [mailing list](#). You can also directly contact the [project coordinator](#) anytime!